

Recommendation stream by grain

Y and X . One row is one order. Y is conversion on that order — purchase or not. It is the outcome only: never a feature, never a ranking key, never a subset key. X is the serving table, sliced the way the log is written.

table	what sits in X	what the row is
order	amount, hour, items, channel	this ticket
merchant	category, GMV, SKU count	the store on this ticket
user	tenure, past frequency	the buyer on this ticket
dumped all	every column concatenated	do not put this in production

South / north is a merchant attribute, not Y . The batch index only marks a quiet reference window versus new traffic — it is not a treatment.

How the marketplace actually logs. A recsys serving log is three tables that share a foreign key, not one embedding soup. The order table is this ticket: amount, hour, items, channel. The merchant table is the store on that ticket. The user table is the buyer. Those three clocks are not the same clock. Ticket size and channel mix move when south stores start selling a different basket. Merchant GMV is last window's ticket size for that store — not this ticket, not conversion. User frequency is a closed-period snapshot — not this ticket, not conversion, and not a running clock. Tenure is age at the start of the stream, not a function of this conversion. Tenure and past frequency move only if the buyer panel itself walks. Users shop across south and north, so a south-store mix shift does not show up as a user-feature incident. Dump every column into one table and a quiet user grain is drowned by a loud order grain: the frozen predictor looks broken before anything real has happened. The working rule is: watch each table on its own, then join back with merchant id or user id.

Two incidents, three tables. Fit a conversion predictor once on the quiet window. Each new batch is scored against that frozen error floor. A ring means this table is no longer paying rent the way it did on reference.

We look at two business incidents on the same south accounts. The first is a *customer-mix* walk: south ticket size and channel leave the reference panel, and conversion follows because the people in the basket changed — an acquisition / category story. The second is a *conversion* walk: south mix stays put, but purchase given the same basket moves — ranking, creative, or the conversion surface, not growth. Table 1 is the watch on each serving table, written as ops time.

Table 1: Frozen watch on each serving table. “Too early” is a false alarm. “Leak” means mix did not move — conversion came in from the order table; do not page growth.

serving table	south customer mix	south conversion
order	rang with south	rang, mix still
user	leak	leak
dumped all	rang with south	rang, mix still

The order table is the one that actually moved. It rings when south merchants start to walk. The user table never moved on mix — tenure and frequency were never the walk — but the watch can still ring because conversion leaked in from the order table. That ring is not a growth-team incident. The dumped table rings with the order table, because GMV is last window’s ticket size — the same south walk, copied onto extra columns. It is still the wrong clock: a quiet user grain mixed with a loud order grain. Leave it out of production.

Where to look, who to page. After a ring, do not retrain the whole stack. Name the fields on the table that moved, then name the accounts. On the mix walk the order table points at amount, channel; the merchant table points at GMV; the user table recovers nothing — those fields do not live on that table. On the conversion walk the order table still names amount (it sits in the conversion surface) while the mix of X itself has not walked. Table 2 splits the same order table by south versus north.

Table 2: South versus north on the order table. Mix is the basket (ticket size, channel). Conversion is purchase on that ticket. North is the control.

incident	south mix	south conversion	north mix	north conversion
customer mix	moved	moved	still	still
conversion	still	moved	still	still

South is where the walk lives; north stays near reference on both incidents. On a mix walk, south ticket size and channel move and conversion follows — page category ops on those south stores, check the basket, do not open the conversion model first. On a conversion walk, south mix is still and conversion moves — page ranking / conversion on south, do not tell category that “the panel changed.” Users mix across merchants, so the user table cannot tell south from north. Do not page the growth team for a user-feature incident that is actually a south-store mix shift.

What to do next. Do not concatenate the serving log. Watch the order table. When it rings, split south versus north and read mix next to conversion on the same clock. If south ticket size and channel walked and conversion followed, category ops checks those stores first — ranking does not open a model ticket yet. If the same south stores, same basket, conversion moved, ranking checks the south conversion surface — growth is not paged, because the user table never moved. Refresh, then reset the error pool so Monday’s walk is not Wednesday’s alarm. Where to look first: not a unique split of blame across columns.